Chapter 8, Lecture 7

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Confidence intervals

We will discuss three types of confidence intervals.

- Asymptotic Confidence intervals based on the asymptotic normality of the MLE
- Bootstrap confidence intervals
- Exact confidence intervals (possible in some special cases)

This lecture we discuss how to construct asymptotic confidence intervals based on the asymptotic normality $N\left(\theta, 1/\{nI(\theta)\}\right)$ of the MLE $\widehat{\theta}$. So this method works under suitable regularity conditions only.

Let $z(\alpha)$ be the α-upper point of a $N(0,1)$, that is,

$$\alpha = \int_{z(\alpha)}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}x^2)\mathrm{d}x.$$

A little algebra shows that

$$\begin{aligned} 1-\alpha &\approx \mathbb{P}_\theta\left\{-z(\alpha/2) \le \sqrt{nI(\theta))}(\widehat{\theta}-\theta) \le z(\alpha/2)\right\} \\ &\approx \mathbb{P}_\theta\left\{-z(\alpha/2) \le \sqrt{nI(\widehat{\theta}))}(\widehat{\theta}-\theta) \le z(\alpha/2)\right\} \\ &= \mathbb{P}_\theta\left\{\widehat{\theta} - \frac{z(\alpha/2)}{\sqrt{nI(\widehat{\theta}))}} \le \theta \le \widehat{\theta} + \frac{z(\alpha/2)}{\sqrt{nI(\widehat{\theta}))}}\right\} \end{aligned}$$

Our $1-\alpha$ asymptotic confidence interval for θ is thus

$$\widehat{\theta} \pm z(\alpha/2)\frac{1}{\sqrt{nI(\widehat{\theta})}}$$

- We give not only a point estimate $\widehat{\theta}$, which we know is close but likely not exactly equal to θ, but give an interval, hopefully small, that contains with probability $1-\alpha$ the ture parameter value.
- The interval is symmetric around the estimate $\widehat{\theta}$
- The term $1/\sqrt{nI(\widehat{\theta})}$ in the width of the interval is the asymptotic standard error of $\widehat{\theta}$.
- The width tends to zero as $n \to \infty$, which is desirable of course. We prefer shorter confidence intervals.

Examples

(1) Exponential(λ). We found that $\widehat{\lambda} = 1/\overline{X}_n$ and the asymptotic variance (AV) equals λ^2/n. Hence our $1-\alpha$ asymptotic C.I. for λ is

$$\widehat{\lambda} \pm z(\alpha/2)\frac{\widehat{\lambda}}{\sqrt{n}}$$

See additional simulations in R.

(2) Hardy-Weinberg. Recall that

$$\widehat{\theta} = \frac{X_2 + 2X_3}{2n}$$

which is approximately $N(\theta, \theta(1-\theta)/(2n))$. Hence our asymptotic C.I. for θ is

$$\widehat{\theta} \pm z(\alpha/2)\sqrt{\frac{\widehat{\theta}(1-\widehat{\theta})}{2n}}$$

Using the numerical example with $x_1 = 342, x_2 = 500, x_3 = 187$, we get $\widehat{\theta} \approx 0.4247$ and for $\alpha = 0.05$, $z(0.025) \approx 1.96$ resulting in the 95% interval $(0.403, 0.447)$. See additional simulations in R.